Volume of prisms and cylinders

1 1728 cm^3

2 3 cm

3 9 cm

4 6 m

5

a 48 cm^3 b 32 cm^3 c 217.5 cm^3 d 330 cm^3

e 32 cm^3 f 270 cm^3 g 756 cm^3 h 63 cm^3

i 1800 cm^3 j 187.738 cm^3 k 399 cm^3

6

a No b No c No d Yes

7

a 8 cm b 768 cm^3

8 59.7 cm^3

9 2309.07 cm

10

a 294.5 cm^3 b 307.9 cm^3 c 70.7 cm^3 d 816.8 cm^3

e 263.9 cm^3 f 1608.5 cm^3 g 1057.9 cm^3 h 755.9 cm^3

i 940.1 cm^3 j 883.6 cm^3 k 491.5 cm^3

11

a 552.92 cm^3 b 2211.68 cm^3 c 15.5 L

12 829.38 cm^3

13

a 131.43 cm^2 b 1445.73 cm^3

Additional questions

14 $\mathrm{r} = 2.3\ \mathrm{cm}$

15

a $\mathrm{Volume} = 400\pi\ \mathrm{cm}^3$

b $\mathrm{height} = 20\ \mathrm{cm}$

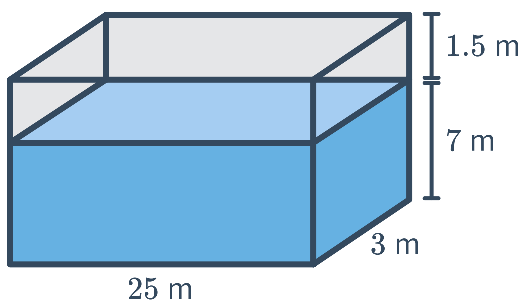
16 8 die

17

a $183.6\ \mathrm{cm}^3$

b $156.06\ \mathrm{cm}^3$

18 Any dimensions where the water occupies $524.6\ \mathrm{m}^3$ of space with an extra $1.5\ \mathrm{m}$ clearance at the top of the tank. For example,



19 Either option will fit the volume of all of items. The cost of purchasing $15\times$ Box 1 is slightly cheaper than the cost of $36\times$ Box 2 so if you want to save money you can select the option to buy $15\times$ Box 1.